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Intelligent Decision Support System for Smart Agriculture

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prediction to determine water retention ratio, soil nutrition ratio so that the accuracy of the system can be increased. Geographic location based customized fertilizer, pesticide and watering need prediction can be used to make suitable predict for specific region and weather.

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It is rightly said by researchers that Indian poverty rate can be decreased three times faster by investment in only one field i.e. agriculture rather than investing in IT, infrastructure, education etc.. It is estimated that by next 30 years, the agriculture with the help of Internet of Things will increase food production by seventy percent and will be able to feed up to 9.6 billion people. The challenges of rising climate change and extreme weather conditions has to be set against the demand for more food, with the natural impact affected by exhaustive farming practices. Thus appropriate water supply is a necessity in agriculture; crops can be either soiled by either water surplus or scarcity. Proposed system has capability to save over 189 billion liters of fresh water in a year. There are intelligent pest Management and Control systems that can monitor pests remotely. This would propose farmers what type of pesticide to be used and in exact quantity. The farmers can now seek the fruits of his yields and protect his fields with no worrying about precision. Sometimes farmers are not even aware of how much pesticide to be used, which will result in over usage of chemicals and hence making it toxic for human utilization. Farmers can now use pest control sensors such as acoustic sensor to predict pest behavior. Proposed architecture will be helpful in controlling the damage done by pests on a large scale by the use of acoustic sensor.

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Intelligent Decision Support System for Smart Agriculture

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Abstract— Indian economy is mainly dependant on agriculture. Thus development in this field will highly contribute to the economic welfare. A large amount of uncertainty exists in agriculture during prediction of fertilizer, pesticide, watering requirement of every individual crop during various stages of its growth. Thus there is need to develop a system which will help farmer in producing quality and quantity crop production. Fuzzy inference based controller integrated with IOT can be implemented for better precision.

The designed decision support system integrates various environmental factors such as moisture, temperature, soil nutrition level, pest sound frequency and soil PH to formulate and estimate individual crop watering requirement. System is smart enough to start and stop irrigation automatically considering human expert opinion learning through feedback provided. System will also provide alert and recommendation on water soluble fertilizer and pesticide requirement for individual crop. Soil condition can be analyzed for pre processing before prediction to determine water retention ratio, soil nutrition ratio so that the accuracy of the system can be increased. Geographic location based customized fertilizer, pesticide and watering need prediction can be used to make suitable predict for specific region and weather.

Keywords—Dielectric soil moisture sensors; irrigation efficiency; irrigation scheduling; tensiometer; wireless sensor network.

I. INTRODUCTION

It is rightly said by researchers that Indian poverty rate can be decreased three times faster by investment in only one field i.e. agriculture rather than investing in IT, infrastructure, education etc..It is estimated that by next 30 years, the agriculture with the help of Internet of Things will increase food production by seventy percent and will be able to feed up to 9.6 billion people. The challenges of rising climate change and extreme weather conditions has to be set against the demand for more food, with the natural impact affected by exhaustive farming practices. Thus appropriate water supply is a necessity in agriculture; crops can be either soiled by either water surplus or scarcity. Proposed system has capability to save over 189 billion liters of fresh water in a year. There are intelligent pest Management and Control systems that can monitor pests remotely. This would propose farmers what type of pesticide to be used and in exact quantity. The farmers can now seek the fruits of his yields and protect his fields with no worrying about precision. Sometimes farmers are not even aware of how much pesticide to be used, which will result in over usage of chemicals and hence making it toxic for human utilization. Farmers can now use pest control sensors such as acoustic sensor to predict pest behavior. Proposed architecture will be helpful in controlling the damage done by pests on a large scale by the use of acoustic sensor.

Impact of Misuse of agricultural fertilizers on the environment needs to be minimized which has have undoubtedly occurred. It is an inevitable to understand fertilizer and its essential role in high-yielding crops. Crop fertilizer Management can boost crop productivity, which will be a major move and a benefit for farmers. Exact amount and type of fertilizer can be provided to individual plant according to its personal need.

Microwave ovens, train systems, robots, appliances, and many other applications are currently using fuzzy logic controllers. Technology such as Internet of Things combined with agriculture is one such sector, which if collaborated with can result into low-priced yet efficient methods for agriculture, which in turn will give rise to higher quality produce. Proposed approach to agriculture is still underdevelopment and unknown; hence it is high time to take a get into this direction.

Proposed algorithm considers all the former factors and guarantees irrigation at critical depletion. Hence, no water stress occurs. Proposed System uses best possible set of sensors to compute the prediction and recommendation. Water wastage will never take place also opening and closing water gates in an effective manner through farmer's smart phones apps will be a better solution. With the help of sensors attached on the water sluice gates, farmers can able to do a trend comparison analysis.

II. LITERATURE REVIEW

Wireless sensor network has been used in agriculture with the fuzzy logic to support irrigation management at North Italy for vineyard was proposed by F.Viani, M Bertonni and A.Polo considering soil potential water scarcity and rain level parameter[1]. Smart water irrigation monitored by flow sensor for counting no of pulses to track fluid movement and exact estimation of amount of water usage by Pushkar Singh and Sanghamitra Sikia[8].Author Navin Srivastava, Gautam

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Chopra and Prateek Jain has used acoustic Device wireless sensor to acknowledge pest predicted frequency and suitable action can be taken by farmer once frequency crosses a threshold level[2].Decision support model for water irrigation at Iraq using evapotranspiration and soil moisture was modeled by AmthalMousa and Muayad Croock[3].Hygrometer used for soil moisture sensing data and timestamp stored at cloud as CSV file for Data logging by Suraj Kothawade and Abdul Raoof[4].Fuzzy Logic basedirrigation implemented in extreme weather of Qatar [5] with field sensor and radiation calibration as membership function for calculating duration of irrigation. Various pest prediction, analysis and control techniques using wireless sensor network was served by Saeed Azfar and Adnan Nadeem [6]based on their effectiveness and cost. Richard W. Mankin, Clayton W. McCoy and Kathy L. Flanders have mentioned Methods using Acoustic sensor for detection of populations of pest through the sounds or vibrations .This method detect insect pests lying underground [7]. Arduino-is used for smart irrigation having sensors like ESP8266 WiFi module , soil moisture sensor, temperature sensor and water flow sensor is proposed by Pushkar Singh and Sanghamitra Sikia in paper[8]. Sarika Datir and Sanjeev Wagh, has proposed method in monitoring and detection of agricultural disease using Wireless Sensor Network published in International Journal of Computer Applications [9]. IOT Based Automatic Drip Irrigation System using various sensor is implemented and discussed by Ameya Bhale, Suryakant Sawant and Surya Durbha at IIT Bombay [10].

III. PROPOSED SYSTEM

Monitoring plant, soil conditions and weather state together can give attractive returns on investment for farmers utilizing sensing technology. Various sensor usage for capturing soil moisture and nutrimental values, optimal plant growth by controlling water usage , fertilizer necessities for current crop and finally determining custom fertilizer profiles based on soil chemistry are the key feature of proposed model. Symphony Link i.e. Star topologies are an perfect match to place sensors in field, reason being one access point can communicate to a number of sensors 20-100 square kilometers away.

DSS with IOT design for water monitoring, fertilizer and pesticide recommendation approach which supports internet based real time data (environmental factors) collection to formulate, estimate and automate individual crop watering and nutrition requirement considering human expert opinion learning through feedback provided.

A. Proposed Architecture

Proposed system architecture is shown in Figure1.with two main modules preprocessing and prediction. Data collection phase collects data from sensor about crop, soil and weather. In preprocessing various soil and weather parameters are used for further complex computation. Prediction phase involves algorithm for prediction of three main components for plant growth i.e. fertilizer, water and pesticide.

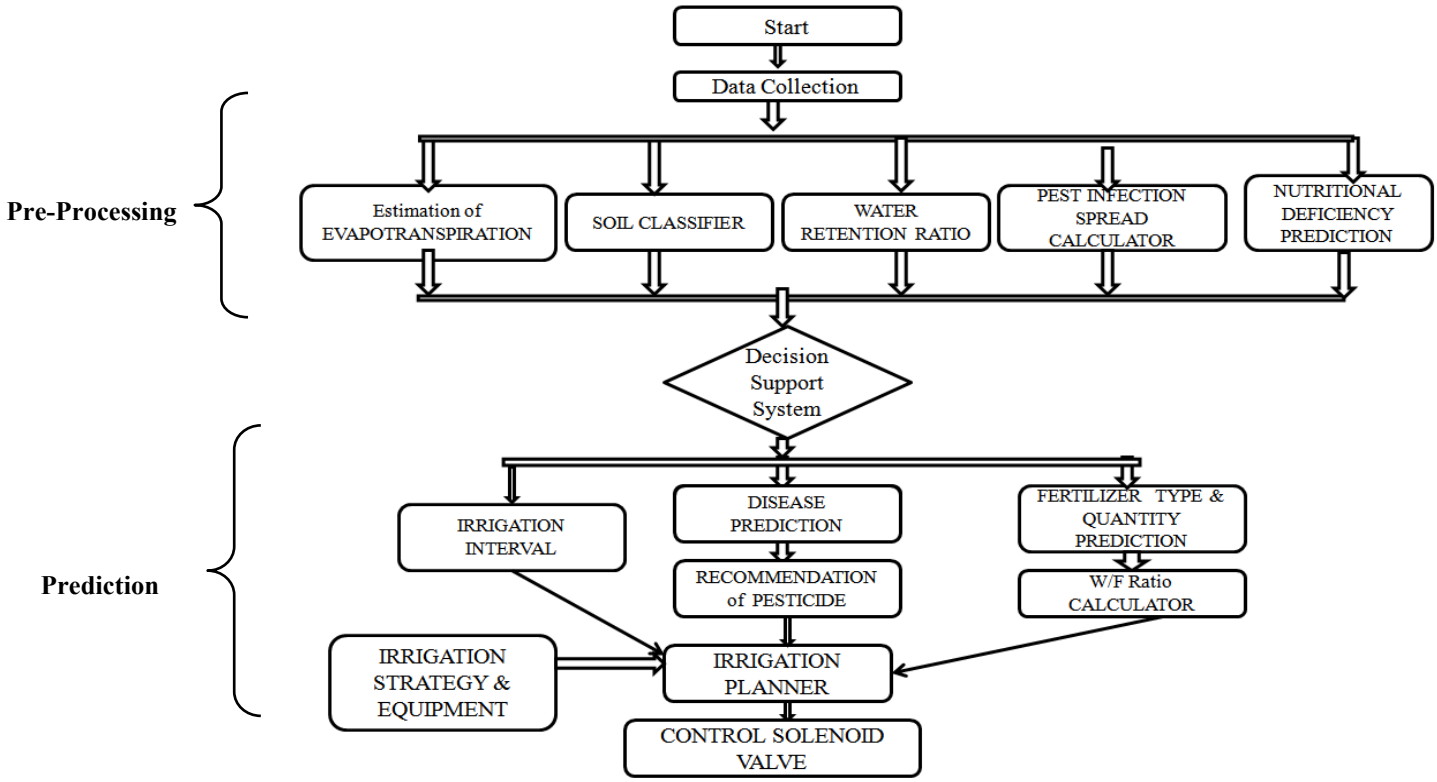

Fig. 1. System Architcture.

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1) Preprosessing: Collected data is preprocessed using computational maodel before sending it to decision support system.

a) Estimtion of EVAPOTRANSPIRATION:

ET is a critical parameter which determines the quantity of water to be provided through irrigation. Various climatologically parameters should be considered when evaluating reference evapotranspiration (ETo),[] which are: air temperature, sunlight and heat index. Thornthwaite's Method equation is considered as a standard method for determining evapotranspiration shown in equation [1].

$$ET=1.6\ La\ (10T/I)\ 2 \quad (1)$$

Where,
ET = Evapotranspiration
L_a = Number of Hours of Sunlight and days in a month
T = Mean monthly air Temperature in degree Celsius
I = Total of 12 monthly values of heat index i.

b) Soil Classifier:

Estimation of the soil water content is a complex task since the Properties of the soil are highly variable in space and a large set of experiments are required to properly characterize the soil response to rain and irrigation. In this sense, a finite set of predefined soils already characterized in the state of the art have been considered as a-priori information of a soil classifier using USDA soil classification used to estimate the soil water quantity[1].

c) Water Retention Ratio:

Dielectric sensor woks superb in very homogeneous soil media like sand or rock wool. However, mostly they require a soil specific calibration and are inclined to soil heterogeneity. Excess or gravitational water drains quickly from the soil after a heavy rain because of gravitational forces (saturation point to field capacity). Crop mostly use very little amounts of this water before it moves out of the root zone. After the excess water has drained, available water is preserved in the soil (this is known as field capacity to wilting point). This water is the highly important for crop production. Plants can use almost 50 percent from it without having stress. Unavailable water is soil moisture which is seized so tightly by the soil that it cannot be extracted by the plant. Water availability is illustrated in the figure 2 by water levels in three different soil types.

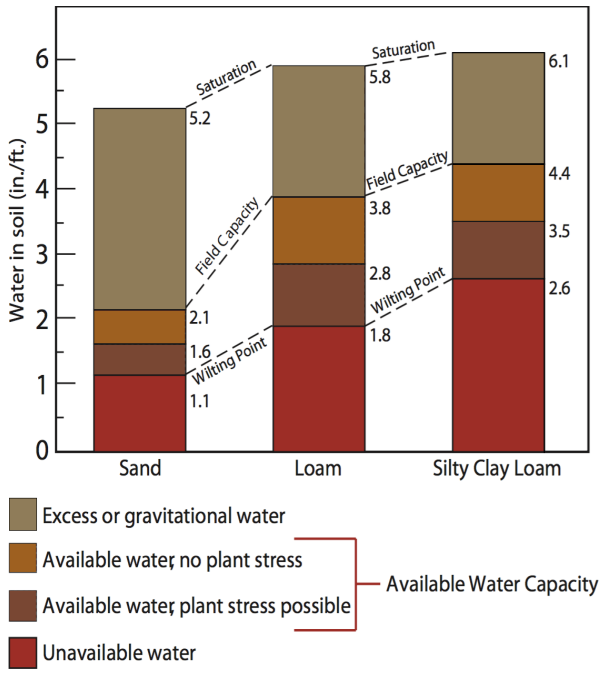

Fig. 2. Water Retention Ratio Claculation

d) Nutritional Deficiency Predictior:

The nutrients present in the soil will change over a period of time thereby reducing the yield/crop. NPK sensor can calculate Nitrogen, Potassium and Phosphorus level in the soil. Soil PH, NPK level, crop growth can be considered to decide nutritional deficiency of soil which will further help in decide water soluble customized fertilizer recommending.

2) Decision Support System:

Human experience is utilized in designing the system model is made possible by the use of fuzzy logic, which is used to frame the solution to a problem in easily understandable manner. Parameters from preprocessing values will act as membership values which will be applied to membership function to get the alpha-cut set. Neuro Fuzzy Rule base will be fired to get output rule. Output rule is then cut by alpha cut-set to get several trapezoidal shapes. The defuzzification is done using Mamdani Suzeno 3Center of Gravity over the area to decide is there need to irrigate. Mamdani Suzeno model can be used due to its suitability to optimization, adaptive techniques, and mathematical analysis.It is given by the formula in equation [2]:

$$Y = \frac{\sum_{i=0}^{n}(Wi*Zi)}{\sum_{i=0}^{n} Wi} \quad (2)$$

Where Wi i_s the output membership value of every rule, Zi is a weight coupled with every rule, N is total number of rules in rule base, and Y is the defuzzified output. If irrigation is required then exact quantity will be predicted by fuzzy controller.

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3) Irrigation Planner:

After deciding exact quantity of water for irrigation, entire water quantity can be watered at one stretch. Irrigation planner decides the schedule for watering the decided quantity. Schedule consists of interval for irrigation and quantity based on the knowledge of irrigation strategy and equipments used i.e.drip irrigation or sprinkler irrigation. Output of irrigation planner is then given to relays to control solenoid valve positions. Figure 3 elaborates the irrigation planner working using Neuro Fuzzy Controller shown in Figure 3.

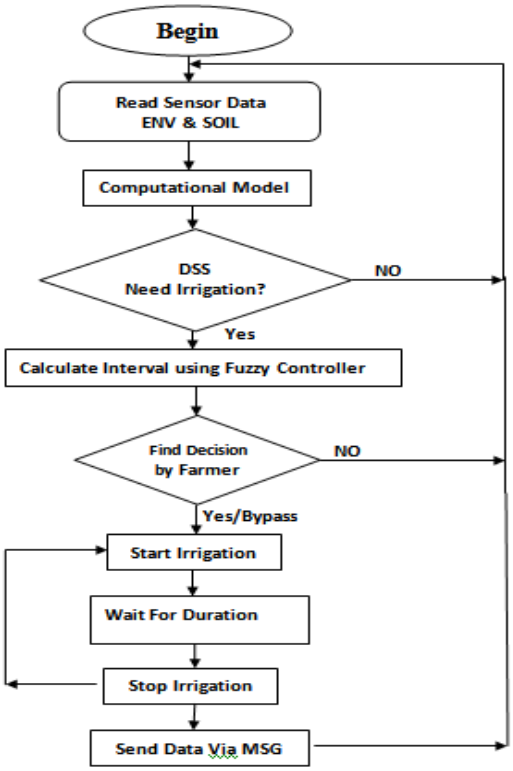

Fig. 3. Irrigition Interval Planner.

4) Dieases Prediction and Recommendation:

Pests can be monitored remotely by intelligent monitoring systems. These systems are capable enough to propose farmers what type of pesticide to be used and in exact quantity. The farmers can now seek the fruits of his yields and protect his fields with no worrying about precision. Sometimes farmers are not even aware of how much pesticide to be used, which will result in over usage of chemicals and hence making it toxic for human utilization. Farmers can now use pest control sensors such as acoustic sensor to predict pest behavior. Proposed architecture will be helpful in controlling the damage done by pests on a large scale by the use of acoustic sensor.The input membership function to identify particular pest is by its frequency spectral template, time of occurrences and geographic distribution.

*a) Pest Training:*Pest generally generate specific sount for 2 to 300ms to communicate with each other. They make noise when they make movement or while feeding.100 to 200 such sounds can be averaged to find sound frequency spectral template of each pest or insect. Other parameter for pest prediction training is genration of probability distribution function for time of occurance of pest. The probability of detecting a sound by a specific species of insects and at a precise time of a day is recorded in Probability distribution function which is an array denoting these values. The sound frequency of insects can be recorded by acaustic. Third membership function is the geographic distribution of the species of interest.This data can be collected from historical records, from relevant literature or simply from field technicians or biologists.

b) Pest Detection Algorithm: In real time recorded signal from acaustic sensor is acting as a input to pset detection algorithm. Use a 0.01 sec long sliding window on amplified recorded input real time signal. If the fundamental frequency is within the range then save the occurrence time of the current window.Remove noise present in the interested part of recording. Extract truncated frequency spectrum.Move the sliding 0.1 sec forward. Repeat the procedure till you reach the end of recorded signal. Given the geographic location where the insect sound was intercepted, compute the probability of observing an insect from class at that specific location using the graphic distribution of species shown in Figure 4.

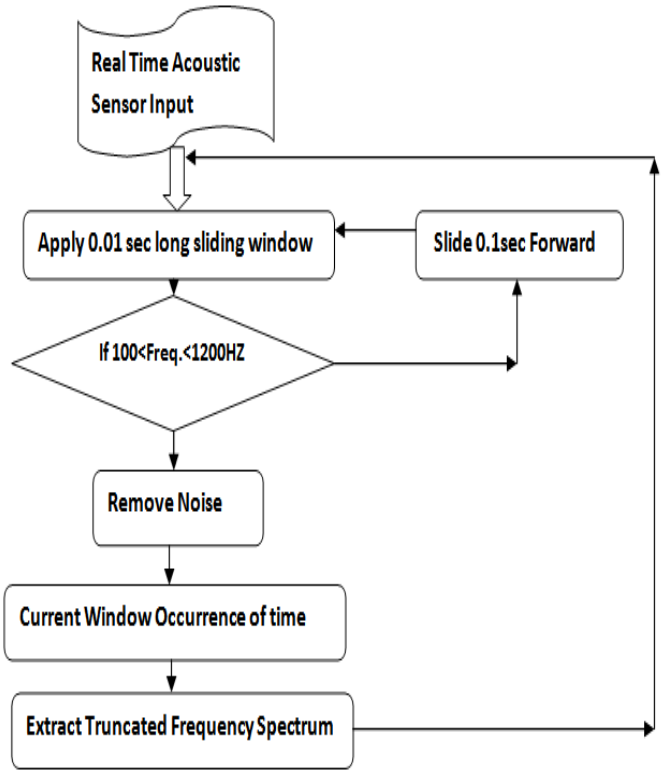

Fig. 4. Irrigition Interval Planner.

*c) Insect Classification:*Use the fuzzy Controller to classify an unknown insect *.use FL based* Classification that

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classifies an object to its most probable class.firstly identifing Time of occurances time of insect sound.This can be achieved by obtaining the occurrence time of each sound from the detection results. The probability of detecting a sound by a specific species of insects and at a precise time of a day is recorded in Probability distribution function which is an array denoting these values. So once a time is given, one can simply check the array to get the probability.Insect Geographic Distribution by identifying Spread of Diease on various geographical Location of Diease. And then defuzzifying the values to get the correct insect classification.

IV. RESULT AND DISCURSSION

Input membership function for water need prediction is temperature, soil moisture and humidity. Input membership function for pest prediction is recorded sound frequency, geographic location and occurrence of time. In the same way Input membership function for custom fertilizer prediction is soil NPK value ,crop type and soil PH value. Few examples of membership function are in Figure 5 which shows input membership function for calibration of weather temperature.

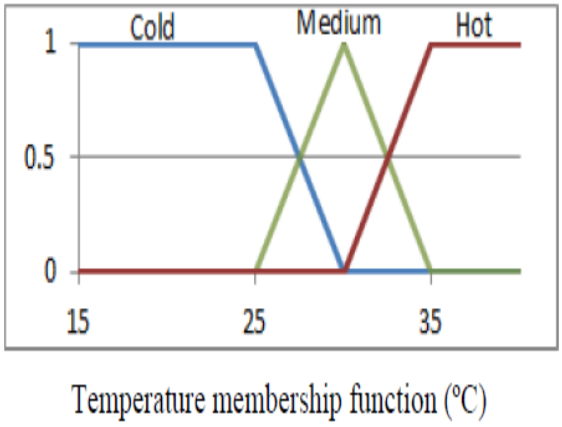

Fig. 5. Input Membership Function(Temprature)

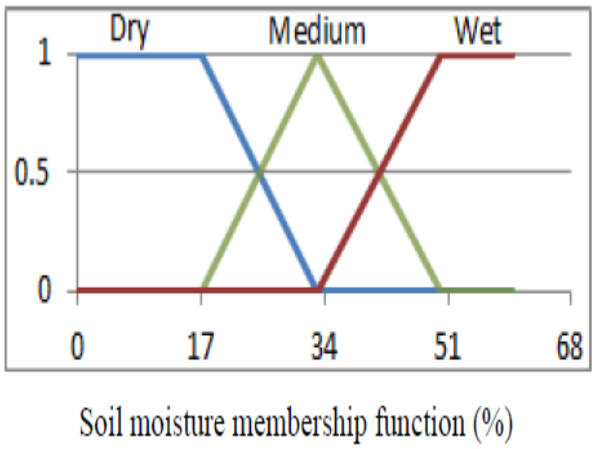

Fig. 6 Input Membership Function(Soil Moisture)

Figure 6 shows input membership function for soil Moisture. Figure 7 demonstrates output membership function for duration for watering through fuzzy inference engine.

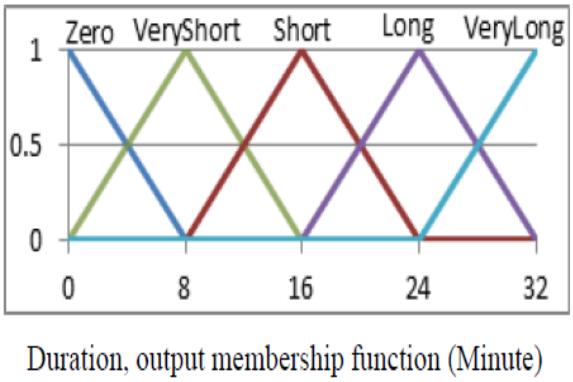

Fig. 7. Output Membership Function(Watering Duration)

Figure 8 shows the graphical user interface of Agro-Vendor Application in farmers smart phone which would help in increasing the probability of getting good deal for their product. Also the weather and soil data at regular interval will be send on cloud to made it available for further research and survey purpose. Farmer and vendor will be users of this application interface Farmer can register, send product data, upload real time soil data, see vendor quotation and select the best deal. Agro-Vendor can get notification from various farmer about their product .After analyzing it with market survey he can send quotations to farmers.

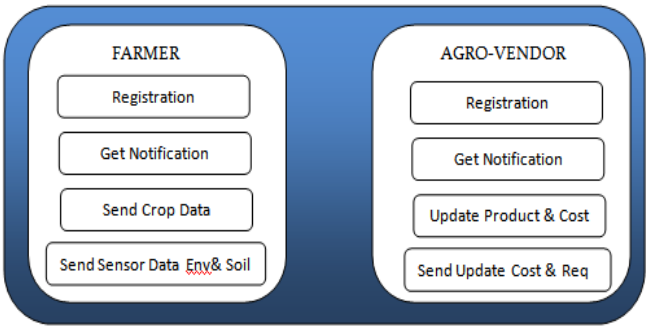

Fig. 8 AGRO FARMER VENDER APP(Features)

IV. CONCLUSION

Recommendation of water soluble pesticide, fertilizer and water need prediction for individual plant. Efficient water management is innovative method. Proposed system is easy to use comparatively low cost, automated water sector -flow rate measuring and encourage water conservation. Proposed system is scalable with input membership functions.

Propose system can be expanded further for bill generation sending bill via messages also water theft detection can be done. Use the conductivity sensor which will indicate the quality of water distributed. Power supply via solar panels can be added to save electricity energy.

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